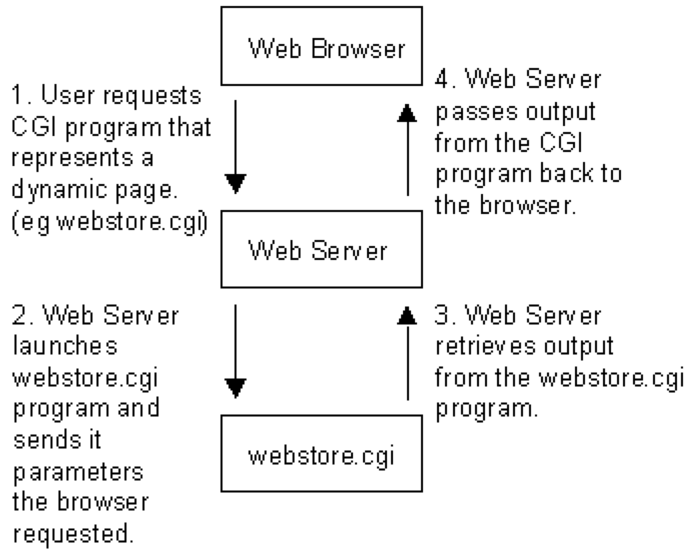
Intro to Webservers-71

1. Introduction

A **web server** is software or hardware that serves content to users over the internet or an intranet. It processes client requests, typically using the **HTTP/HTTPS protocol**, and delivers resources such as web pages, images, videos, or applications. Web servers are the backbone of the modern internet, enabling communication between users' browsers and hosted applications.



2. Functions of a Web Server

1. **Handle Client Requests**
   - Accepts requests from browsers (clients) using HTTP/HTTPS.
   - Routes them to the correct resource.
2. **Serve Static Content**
   - HTML, CSS, JavaScript, images, and videos.
3. **Process Dynamic Content**
   - Works with application servers or interpreters like PHP, Python, or Java.
   - Example: When you log in to Gmail, dynamic content is generated.
4. **Manage Connections**
   - Handles multiple requests simultaneously using multi-threading or asynchronous processing.
5. **Implement Security**

- Provides HTTPS encryption (SSL/TLS).
- Supports authentication, authorization, and access control.

6. **Logging and Monitoring**
    - Records access logs, error logs, and performance metrics.

3. Common Web Servers in Use

1. **Apache HTTP Server**
    - Open-source, widely used.
    - Supports modules for PHP, SSL, authentication, etc.
2. **Nginx**
    - Lightweight, high-performance.
    - Often used as a reverse proxy or load balancer.
3. **Microsoft IIS (Internet Information Services)**
    - Windows-based.
    - Integrated with ASP.NET applications.
4. **LiteSpeed**
    - Focuses on speed and scalability.
    - Known for efficient handling of high traffic.
5. **Tomcat** (specialized for Java)
    - Runs Java Servlets and JSPs.
    - Often used with enterprise Java applications.
6. **Caddy**
    - Modern, simple configuration.
    - Built-in HTTPS support.

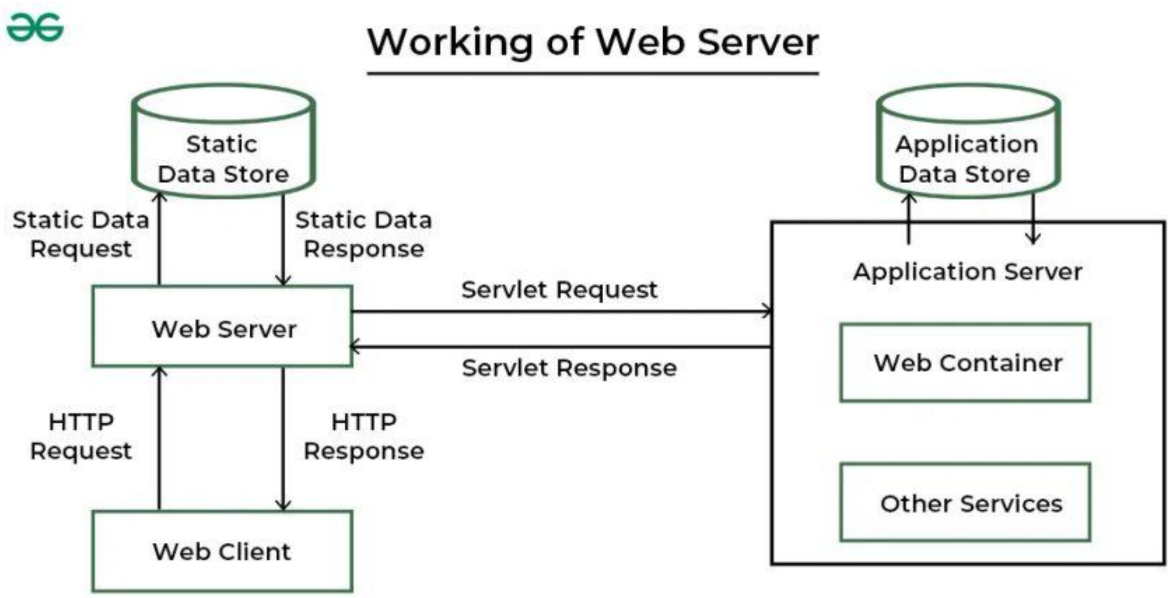
4. How a Web Server Works

1. User enters a URL in the browser.
2. Browser sends an HTTP request to the server.

3. Web server receives the request and checks for the requested resource.
   - If it is static, it serves the file directly.
   - If it is dynamic, it forwards the request to an application server or script engine.
4. The server sends back an HTTP response with status codes (200 OK, 404 Not Found, etc.).
5. Browser displays the content to the user.

5. Types of Web Servers

1. **Static Web Server**
   - Serves only static files like HTML, CSS, JS.
   - Fast and simple, but limited functionality.
2. **Dynamic Web Server**
   - Works with back-end logic (databases, APIs, application servers).
   - Example: E-commerce sites, social media platforms.
3. **Reverse Proxy Web Server**
   - Acts as a gateway, forwarding client requests to backend servers.
   - Provides load balancing and caching.



6. Key Features of Web Servers

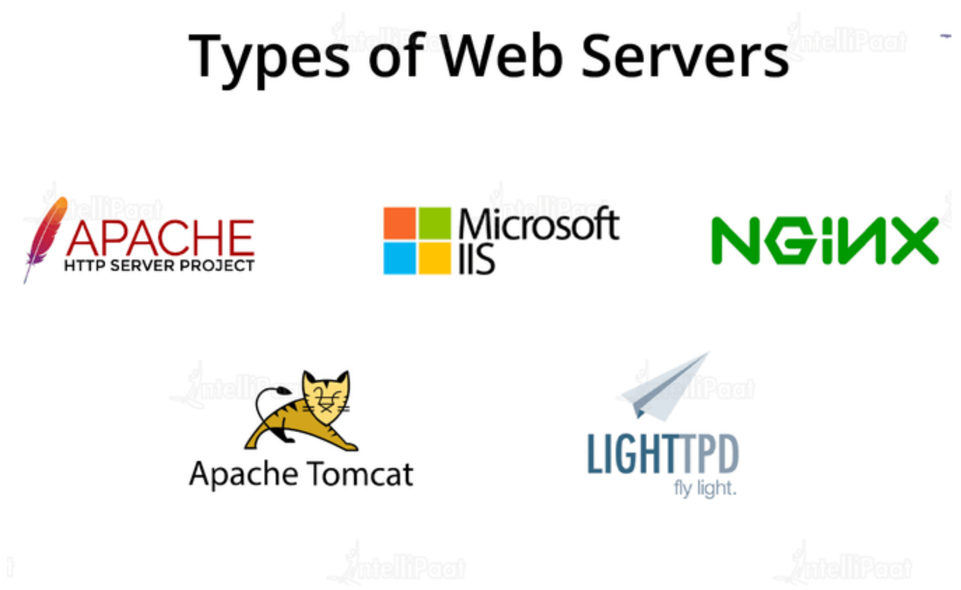
- Virtual hosting (host multiple websites on one server).
- Load balancing to distribute requests across servers.
- Compression (e.g., Gzip) to improve performance.
- Cache mechanisms to reduce load times.
- Support for multiple programming languages and frameworks.

7. Importance of Web Servers

- Enable online presence for businesses and individuals.
- Provide platforms for e-commerce, education, entertainment, and communication.
- Serve as entry points for applications and APIs.
- Support scalability and high availability in cloud environments.

8. Examples in Real World

- **Apache** is used by WordPress websites.
- **Nginx** powers large-scale sites like Netflix and Dropbox.
- **IIS** is used in many enterprise Microsoft environments.
- **Tomcat** supports enterprise Java applications like banking systems.



9. Challenges in Web Servers

- Handling high traffic during peak usage.
- Preventing cyberattacks like DDoS, SQL injection, or XSS.
- Ensuring uptime and disaster recovery.
- Optimizing performance for global users.

Web servers are essential for delivering content and applications over the internet. They provide speed, scalability, and security for both static and dynamic content. Understanding web servers is the foundation for web development, system administration, and cloud computing.